cyber security

Vulnerability Assessment Report For A Live Website

Vulnerability Assessment Report

Target Website: http://demo.testfire.net

Assessment Type: Web Application Vulnerability Assessment

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1.Executive Summary

This report presents the results of a vulnerability assessment conducted on the web application demo.testfire.net. The purpose of the assessment was to identify security weaknesses that could potentially be exploited by attackers.

The assessment was performed using industry-standard tools including Nmap and OWASP ZAP. Several vulnerabilities were identified during the assessment, including SQL Injection risks, missing security headers, and information disclosure issues.

I.Scope of Assessment

The vulnerability assessment was conducted on the following target:

Target Websitehttp://demo.testfire.net

The assessment focused on identifying common web application vulnerabilities such as:

SQL Injection

Security header misconfigurations

Information disclosure

3.Tools Used

Tool	Purpose
Nmap	Network scanning and service detection
OWASP ZAP	Web vulnerability scanning
Canva	Report design

4.Methodology

The vulnerability assessment followed a structured approach:

1. Reconnaissance

Initial reconnaissance was performed to gather information about the target system using Nmap.

I. Vulnerability Scanning

The web application was scanned using OWASP ZAP to detect potential vulnerabilities.

3. Analysis

Detected vulnerabilities were analyzed and classified based on severity.

4. Reporting

All findings were documented along with their risk levels and remediation recommendations.

5. Network Scan Results (Nmap)

A network scan was conducted to identify open ports and running services on the target system.

Command Used: nmap –sV –T4 demo.testfire.net

Observation:

The web server is accessible over HTTP and exposes service information.

Risk Level: Low

Recommendation Disable unnecessary port exposure and hide server version information.

6. Risk Summary

Risk Level	Number of Findings
High	1
Medium	3
Low	1

7. Recommendations

To improve the security of the web application, the following actions are recommended:

* Implement input validation and prepared SQL statements

* Add security headers such as CSP and X-Frame-Options

Use secure cookie attributes
Hide server version information
Implement CSRF protection mechanisms
Conduct regular vulnerability assessments

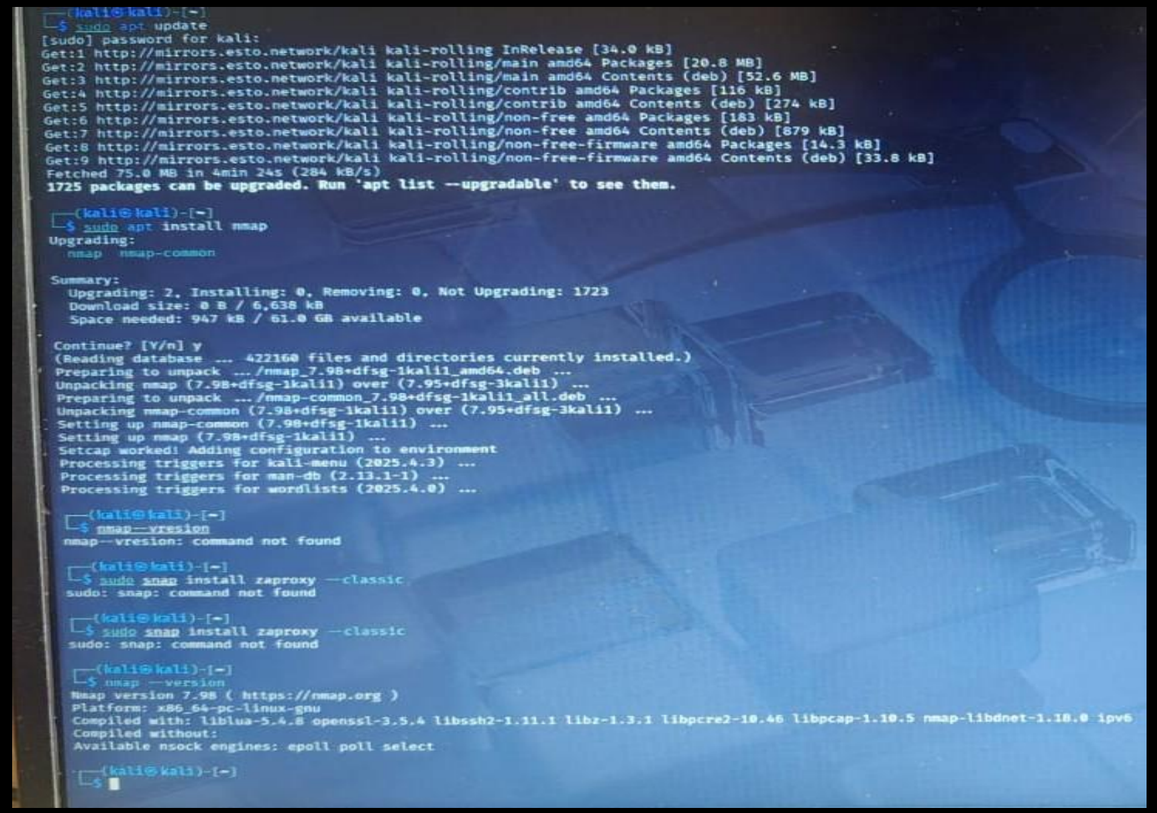
8. Conclusion

The vulnerability assessment identified several security weaknesses within the web application. While most vulnerabilities were classified as medium risk, the presence of SQL Injection vulnerabilities represents a significant security threat.

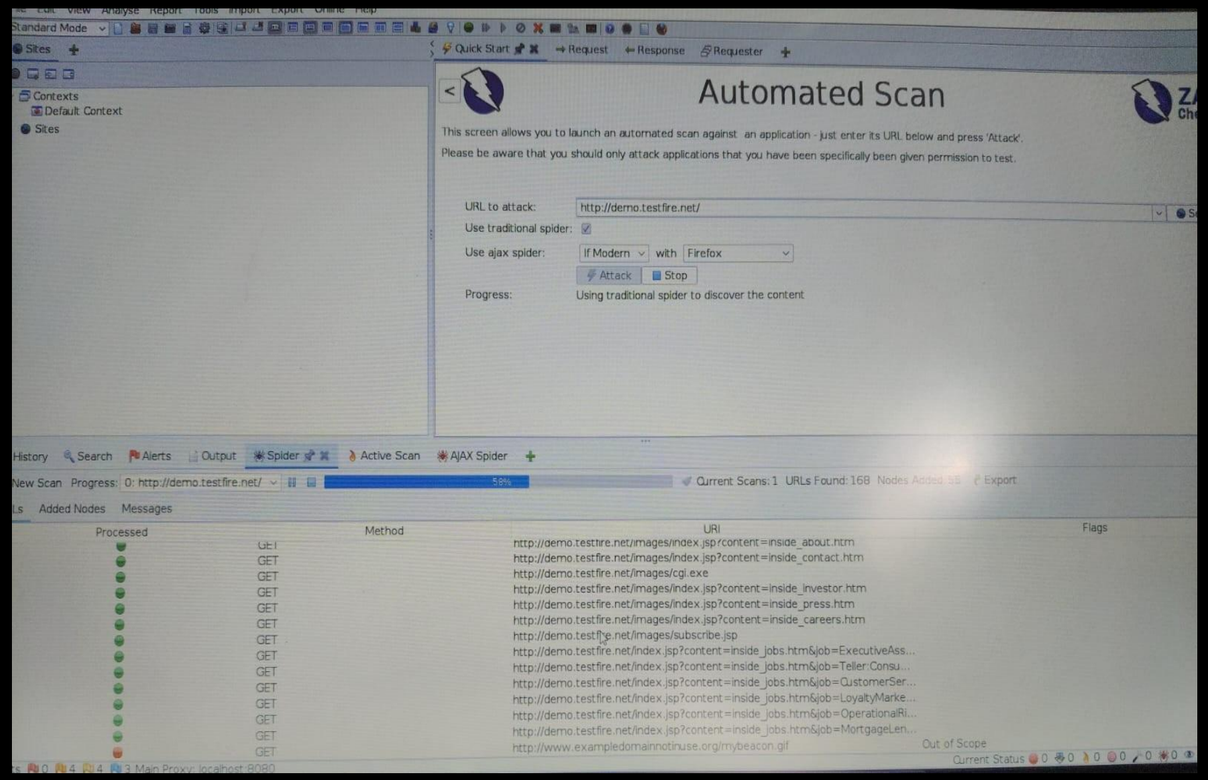
By implementing the recommended security controls, the organization can greatly reduce the likelihood of successful attacks and improve the overall security posture of the application.

C.Appendix

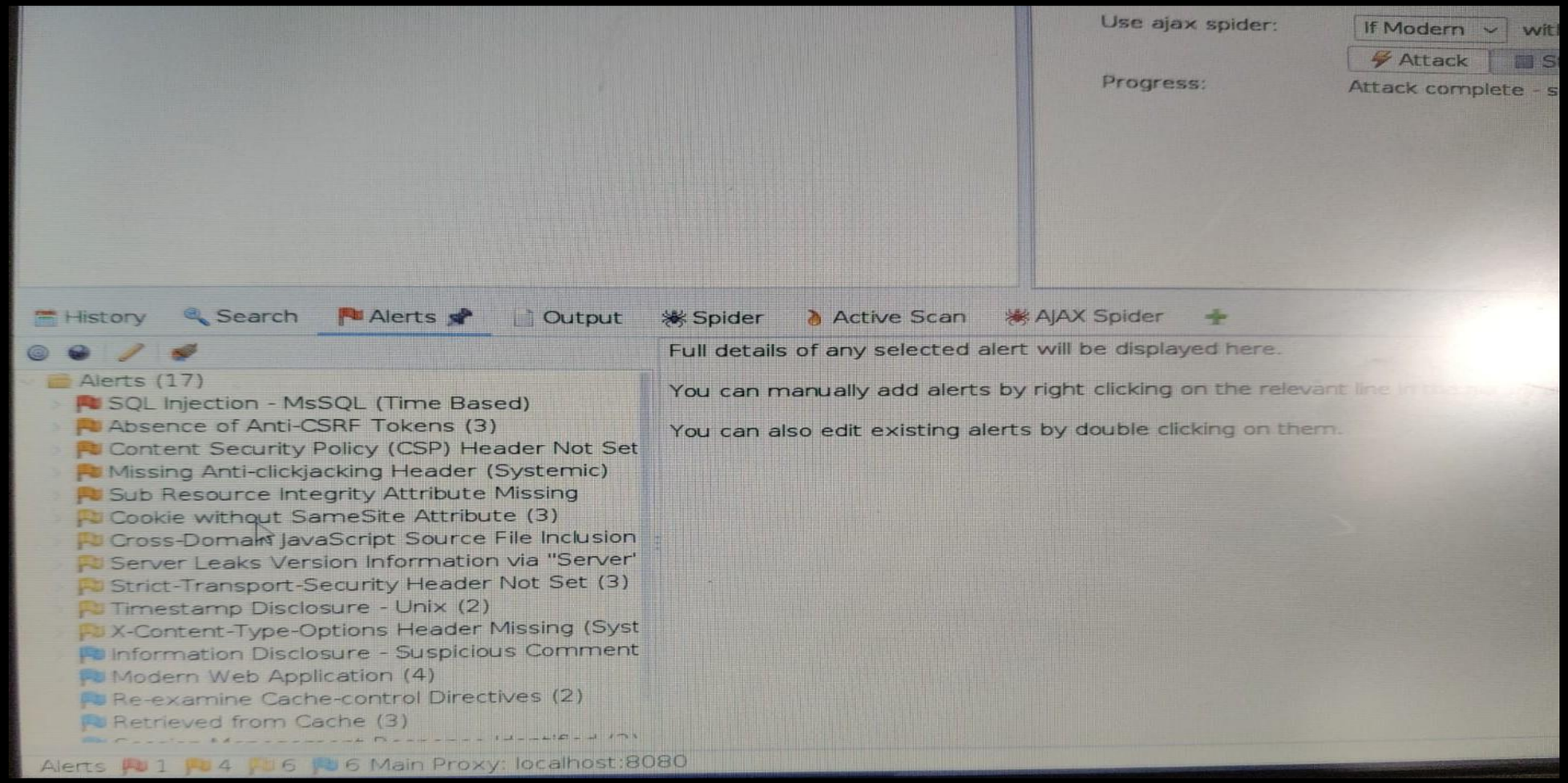
1.Nmap scan results



l.OWASP ZAP active scan



3.OWASP ZAP alerts panel



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THANK YOU!